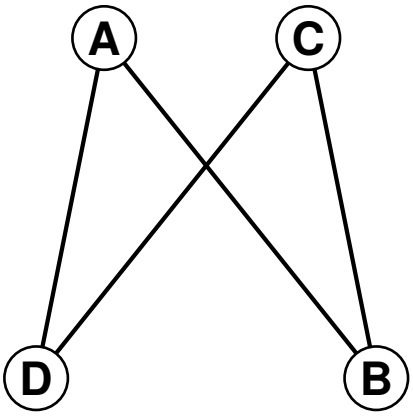
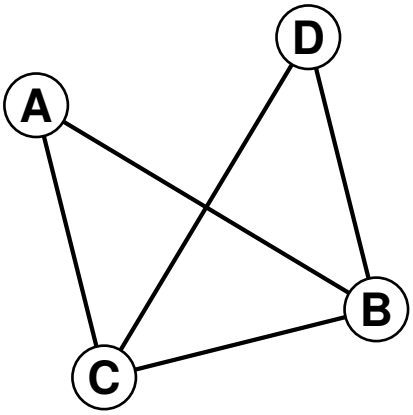
Move the dots and bend the links however you like. Each link stays joined to its own two dots and never lies across a dot. A crossing is a place where one link lies across another.

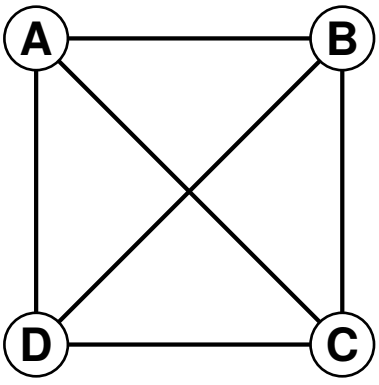
Problem 1: Make each network with dots and pipe cleaners. Move the dots until there are no crossings, then draw what you made in the box.



A – B
A – D
B – C
C – D

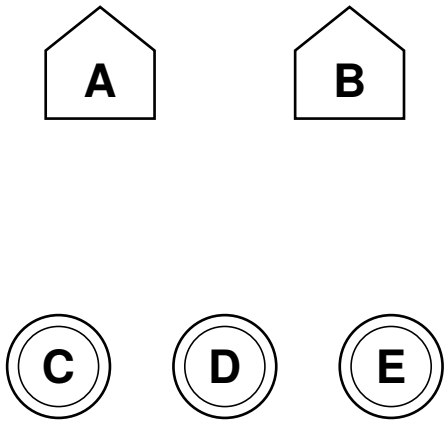


A – B
A – C
B – C
B – D
C – D

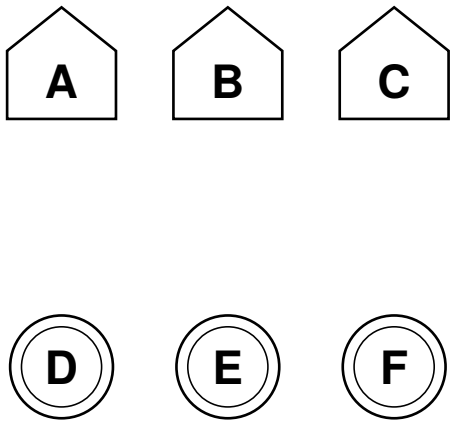


A – B
A – C
A – D
B – C
B – D
C – D

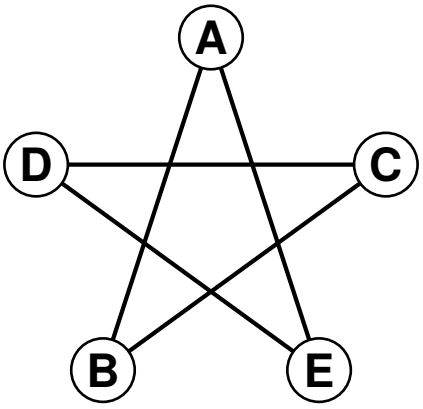
Problem 2: Join each of these 2 houses to each of the 3 wells with pipe cleaners. Can you do it with no crossings? Draw what you made.



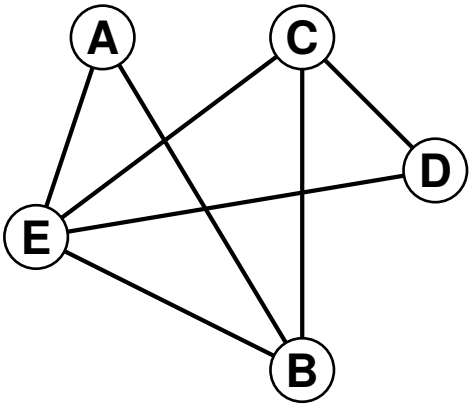
Problem 3: Join each of these 3 houses to each of the 3 wells with pipe cleaners. Can you do it with no crossings? Draw what you made.



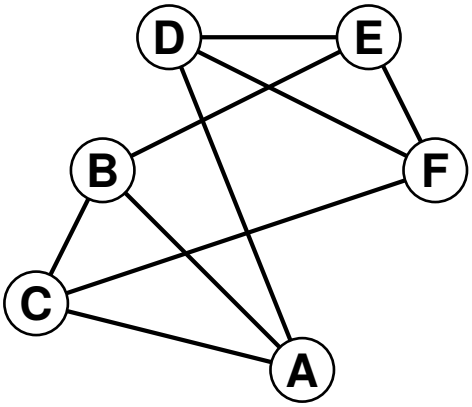
Problem 4: Make each of these networks and move the dots until there are no crossings. Draw what you made.



A – B
A – E
B – C
C – D
D – E

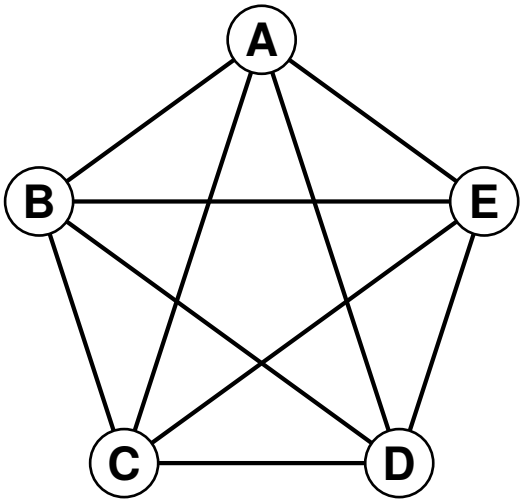


A – B
A – E
B – C
B – E
C – D
C – E
D – E

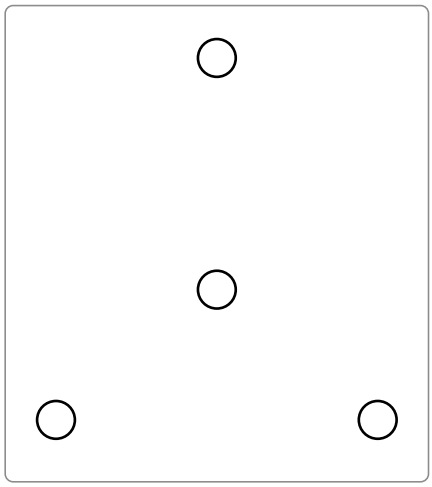
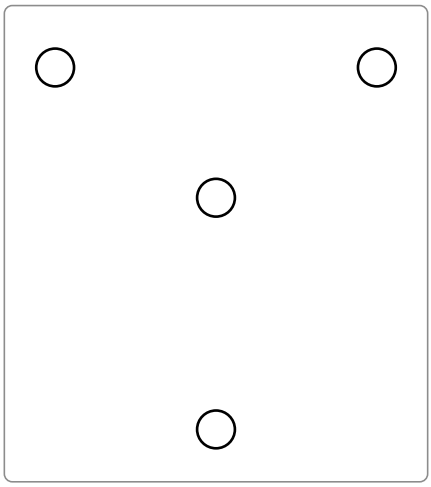
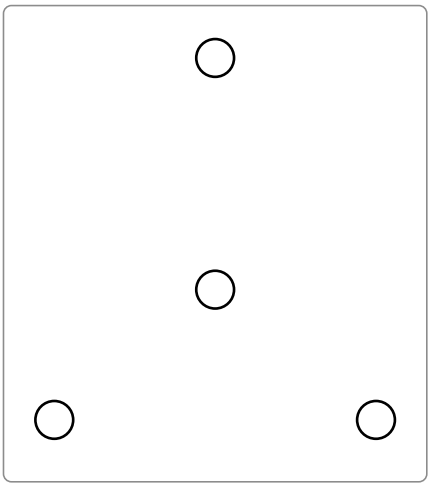
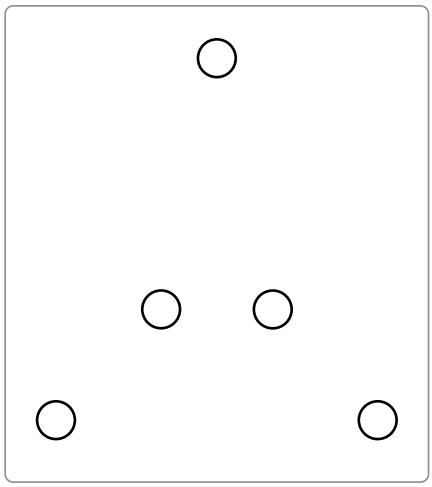
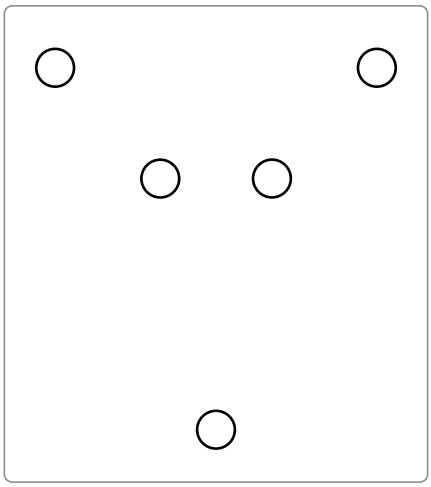
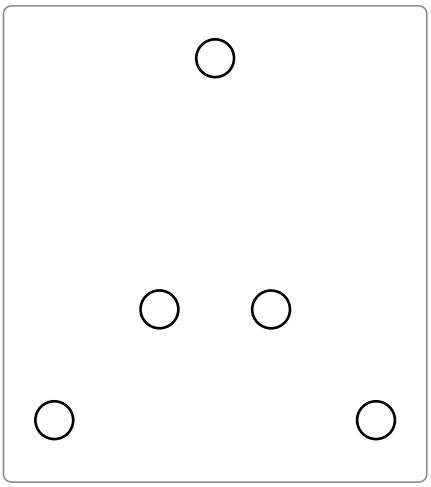


A – B
A – C
A – D
B – C
B – E
C – F
D – E
D – F
E – F

Problem 5: Join each of these 5 dots to every other dot with as few crossings as you can. Draw what you made.



A – B B – D
A – C B – E
A – D C – D
A – E C – E
B – C D – E

Problem 6: Take turns drawing a link between two dots, with no crossings and never two links between the same dots. Whoever cannot draw a link loses. Play with 4 dots, then with 5 dots.

	First player wins	Second player wins
4 dots		
5 dots